A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 13

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.50 x 0.90) x 1.0 **1** = 44.8 t ha^{-1} yr^{-1} (accept 44.7-44.8) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.35 x 0.60) x 0.25 **1** = 5.2 t ha^{-1} yr^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Beans / up-and-down ploughing:** severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall **1** Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable **1** Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land (C = 0.02) needed to bring loss below 1 t ha^{-1} yr^{-1} **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **terracing** → cutting the slope into level steps **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Factors are averages / arbitrary values for typical conditions – rainfall erosivity varies between years / storms **1** Does not include gully erosion, landslides, wind erosion or local variation in soil – based on data from particular regions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	1 mark for the point below. **1** C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s **1** Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (225 - 95) / 225 x 100 **1** = 57.8% (accept 58%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form **1** Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl_2) during the dark winter **1** The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated **1** When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone – levels recover by December as the vortex breaks down **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation **1** Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **correlation** → a relationship in which the two variables change together • **causation** → one variable actually causing the change in the other **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Bacteria in the gut convert nitrates to nitrites **1** Nitrites react with haemoglobin so it cannot carry oxygen **1** Infants are most at risk because they have more of the gut bacteria and their haemoglobin reacts faster – usually from shallow wells receiving arable drainage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrites may be converted in the gut to nitrosamines which are carcinogenic **1** But most research has been on rats / the evidence in humans is not conclusive – nitrites are also ingested in processed food so the source cannot be isolated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Reduce use of nitrate fertilisers / match application to crop need **1** Avoid ploughing / applying fertiliser at times of heavy rain (autumn / winter) **1** Use low-solubility / slow-release fertilisers rather than soluble rapid-release ones **1** Store manure / slurry where it cannot leach or wash into rivers **1** Grow crops with lower nitrogen requirements (eg triticale instead of wheat) **1** Grow legumes with nitrogen-fixing bacteria instead of relying on artificial fertiliser **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **nitrogen-fixing** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Iron(III) sulfate is added to effluent: phosphates are precipitated as insoluble iron phosphate (a fine sediment that can be used as fertiliser) **1 Lakes:** dredging phosphate-rich sediments / adding iron(III) sulfate to reduce phosphate solubility (eg the Norfolk Broads) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 267-268; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 5 points below. **1** Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions **1** Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected – recirculation increases the metal concentration **1** Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) **1** Smelting needs chemical reduction at high temperatures – energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic **1** Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **Leachate** → polluted liquid draining out of waste **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves **1** The vegetation is harvested and incinerated **1** The concentrated metal in the ash is dissolved in acid and separated by electrolysis **OR** decontaminates polluted sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) **1** The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 6 points below. **1 Location:** discharge into the sea / large water bodies where currents dilute and disperse the pollutant **1 Location:** chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations – avoiding valleys and enclosed water bodies where pollutants concentrate **1 Location:** not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers **1 Timing:** restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor **1 Timing:** discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away **1 Timing:** pesticide spraying at night / on still days – noisy operations restricted to daytime – night-flight restrictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** The organisation / person that causes the pollution is responsible for (paying for) the damage caused **1** If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it – works best when penalties are high and escape is unlikely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** It is assumed that the waste will cause pollution / harm until research shows it is unlikely to **1** So release is not permitted until it has been tested – lack of knowledge of a problem is not an excuse / does not reduce responsibility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (60 x 45) / 9 **1** = 300 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (beetles) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Marked individuals mix freely / randomly with the rest of the population **1** No births, deaths, immigration or emigration between samples **1** Marking does not affect survival or behaviour (eg make them more visible to predators) **1** Marks do not wear off / fall off **1** All individuals have the same chance of being caught (traps do not attract or repel marked animals) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Lincoln Index** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	4
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Fewer marked beetles would survive to be recaptured **1** So the proportion marked in the second sample would be too low and the population would be overestimated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Enough time for marked beetles to mix with the population **1** Not so long that births / deaths / migration or loss of marks change the population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen **1** It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted **1** So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) **1** It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mercury is persistent / not degraded – it is an element so cannot be broken down **1** It remains in the sediments and continues to be released / converted and taken up by the food chain – long-lived fish continue to accumulate it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Water current velocity and direction (dispersal) **1 Properties of the pollutant:** solubility, persistence, liposolubility / potential to bioaccumulate **1** Food chain pathways to humans (fish, shellfish) / the critical group **1** Sediment type / adsorption onto clay **1** Volume of the water body / residence time / dilution **1** Sensitive locations downstream (fisheries, drinking water abstraction) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **critical group** → the people most at risk / most heavily exposed • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (15.0 - 2.2) / 15.0 x 100 **1** = 85.3% (accept 85%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Added as an anti-knock agent to smooth combustion and reduce engine wear **1** Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food **1** Lead is a neurotoxin that bioaccumulates – chronic exposure damages the nervous system / brain development in children **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Replacement of lead water pipes / lead solder with copper / tin-based solder – phosphoric acid dosing of water supplies forming insoluble lead phosphate **1** Phasing out of lead-based paint / safer paint removal **1** Lead-free electrical solder **1** Restrictions on lead fishing weights / shotgun pellets **1** Industrial emission controls **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe **1** This prevents lead dissolving into the water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.5	1 mark for the point below. **1** Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable – demand fluctuates daily, weekly, seasonally and short-term – supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants – large-scale electricity storage not yet possible directly so energy must be converted • **Pumped-storage HEP:** proven, rapid response, peak shaving, but limited sites • **Batteries:** lithium-ion for grid and vehicles – factors affecting viability – V2G • **Hydrogen:** electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid – high energy density enables vehicle use • Compressed air with heat recovery – thermal storage (molten salt for CSP, high-volume stores, thermal mass) – flywheels – supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage – energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **HEP** → electricity generated from falling water • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (9 marks)	
	Renewable electricity is intermittent - solar stops at night, wind on calm days, tidal power at slack water - and some of it is unpredictable, while demand fluctuates by the minute, day, week and season, so supply and demand rarely coincide. Without storage, surplus renewable output is wasted and shortfalls must be met by fossil-fuel peaking plants, which defeats the purpose. Electricity itself cannot be stored at scale, so the energy must be converted into another form and back.	
	Pumped-storage HEP is proven and responds within seconds for peak shaving, but suitable sites are few. Lithium-ion batteries now work at grid scale and in vehicles, with vehicle-to-grid schemes turning car batteries into distributed storage, though lithium supplies and lifetime limit them. Hydrogen from electrolysis can be stored, used in fuel cells or injected into the gas grid (power-to-gas), and its high energy density suits vehicles, although each conversion loses energy. Compressed air with heat recovery, molten-salt thermal stores for concentrating solar plants, thermal mass in buildings, flywheels and supercapacitors each fit particular timescales.	
	Storage is not the only answer: a diverse mix of wind, solar, tidal and biofuel smooths supply, smart-grid management shifts demand, and conservation reduces peaks. But these reduce rather than remove the mismatch. Short-term storage can be batteries and pumped hydro; the harder problem is seasonal storage, where only hydrogen or fuels made from surplus electricity look plausible. Without both, renewables can supplement but not dominate a country's supply.	
	Weaker answer - Level 2 (5 marks)	
	Renewable energy is intermittent. Solar panels do not work at night and wind turbines do not work when it is not windy, but people still need electricity then. Storage means the energy can be saved for when it is needed. Pumped storage pumps water uphill when there is spare electricity and lets it flow back through turbines at peak times. Batteries can also store electricity. Hydrogen can be made from electricity and used in fuel cells. So storage is needed for renewables to provide most electricity.	
	What makes the difference	
	• Level 2 explains the mismatch and describes three storage methods correctly, but does not evaluate them (sites, lithium, conversion losses, timescales).	
	• Level 3 distinguishes short-term from seasonal storage - the key insight for 'most of a country's electricity'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 also considers the alternatives to storage (diversity, grid management, conservation) and explains why they are insufficient alone.	
10.2	1 mark for the point below. **1** Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility – DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • **Reducing impact:** restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • **Alternatives:** cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • **Trade-offs:** yield, cost, human health (organophosphates), resistance evolution, pollinator services – risk:benefit – sustainability and the circular economy	25
	• **Judgement:** no pesticide is perfectly specific – combination of property-based selection and integrated control is most sustainable *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • **How properties predict behaviour:** dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • **Control based on properties:** persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones – systemic pesticides and seed coatings • **Heavy metals:** lead uses discontinued, storage at high pH to reduce solubility – mercury chemical form – radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays) – oil properties (floats, insoluble) exploited by booms and skimmers	25

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Question	Marking guidance	Marks
	– density in cyclone separators • **Principles:** CPA and CGM based on properties – emission location / timing – ALARA and BATNEEC – polluter pays – precautionary principle • **New technologies:** satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • **Evaluation:** prevention better than cure – limits of remediation – cost-effectiveness – contrast success of ozone (Montreal) with climate change – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	
	Model answer - Level 5 (22 marks)	
	Pollutants cause damage in proportion to how much reaches sensitive organisms and how harmful it is when it gets there. Both depend on properties, so knowing a pollutant's properties lets us predict where it will go and which control will work.	
	State of matter and density decide dispersal. Solid particulates settle near the source unless they are very small: PM10 and smaller stay suspended, travel far and penetrate lungs, so smoke is controlled at the point of release with electrostatic precipitators, cyclones and bag filters, and smokeless zones prevent its production. Gases disperse widely, and dense gases such as the hydrogen cyanide released at Bhopal hug the ground, so location and emission height matter. Persistence decides whether dilution is acceptable. Sewage and other biodegradable wastes break down, so treatment plus dilution in a river or the sea is sufficient; organochlorine pesticides, CFCs, heavy metals and long-lived radioisotopes are not broken down, so they travel, accumulate and must be contained, banned or replaced - DDT and CFCs were both controlled by substitution once their persistence was understood.	
	Solubility governs the pathway. Water-soluble nitrate leaches into rivers and aquifers, so it is controlled by nitrate vulnerable zones, fertiliser timing and buffer strips; liposoluble pollutants such as DDT and methyl mercury pass into fat and are biomagnified along food chains, as at Minamata, so even low concentrations in water are dangerous and the only effective control is to stop release. Adsorption on clay immobilises caesium and metals, which is why radioactive discharges are monitored in sediments and why activated carbon and polymer adsorbents are used deliberately in effluent treatment.	

Do not write outside the box

Question	Marking guidance	Marks
	Toxicity, specificity and mode of action set acceptable doses and identify species at risk. Neonicotinoids are toxic to insects but not vertebrates because they cannot cross the vertebrate blood-brain barrier, which explains both their popularity and their effect on bees; pyrethroids replaced organochlorines because they are less persistent. For mutagens and carcinogens such as ionising radiation there is no safe dose, so exposure is kept As Low As Reasonably Achievable with shielding, distance, time limits and monitoring. Reactivity explains secondary pollutants - sulfur dioxide oxidised to sulfuric acid, nitrogen oxides and hydrocarbons forming ozone and PANs in photochemical smog - so control targets the primary emissions with flue-gas desulfurisation and catalytic converters. Synergism between smoke and sulfur dioxide, or ozone and sulfur dioxide, means limits must consider pollutants together.	
	Properties interact with the environment: temperature inversions trap pollutants, currents disperse them, pH and oxygen alter solubility. Critical Pathway Analysis combines the two to predict where a discharge will concentrate and to focus monitoring, and Critical Group Monitoring checks the most exposed people. These principles have reduced smoke, sulfur dioxide, lead and organochlorines dramatically in developed countries. Their limits are pollutants with diffuse sources or that underpin economies - nitrate, carbon dioxide - where knowing the properties is not the problem; changing behaviour is. Even there, the precautionary principle applies the same logic: predict the properties of a new substance from similar ones and control it before release, because prevention is always cheaper than remediation.	
	What makes the difference	
	• Every paragraph is organised around a **property**, each linked to a prediction and a named control - the essay answers the question throughout rather than describing pollutants.	
	• Examples span the specification (smoke, acid rain, pesticides, heavy metals, radiation, nutrients, oil), with quantitative and named detail (Bhopal, Minamata, PM10, ALARA), which is what Level 5 breadth means.	
	• The final paragraph evaluates: where the approach has worked, where it has not and why, and reaches a justified conclusion.	
	• A Level 3 version of this essay would describe several pollutants and their effects accurately but only occasionally connect a property to a control, and would end without a judgement.	

Do not write outside the box